

years earlier: that objects move because they are propelled by an external force. No one had challenged this because it seemed common sense. But "common sense" was not enough for Galileo, who believed that theories had to be tested by observation and precise measurement.

Gazing at the stars

In 1609, Galileo became interested in a new invention: the telescope (see box). This changed the course of his research dramatically. He set about building his own improved version, with stronger magnification. With it, he made some surprising discoveries. He saw that the Moon had mountains and craters and that, like the Moon, Venus changed shape in crescent phases. Most startling was his discovery that

"The passage of time has revealed to everyone the truths that I previously set forth."

Galileo Galilei, 1615

Jupiter had its own moons. At the time, it was accepted that the Earth was the center of the Universe, with the Sun, Moon, stars, and planets orbiting around it; that Jupiter's moons were clearly orbiting Jupiter shook this concept to its core. Galileo became convinced that the planets orbit the Sun—a theory that was first proposed by Nicolaus Copernicus in 1543.

Galileo was the first person to use lenses to study the solar system. Jupiter's four largest moons—Io, Europa, Callisto, and Ganymede—are named the "Galilean moons" in honor of his discovery in 1610.

